

☺

P_{m+1} w
(am aratat)

Ți o sy formulă de cuadratură care are gr exact $2m+1$.

$$Q \in \Pi_m, m \geq m+1$$

$$Q(x) - L_m(Q; x_0, \dots, x_m) = \frac{P_{m+1}(x)}{Q_{m+1}} \frac{Q(x)}{Q_{m-m-1}} \Big| w(x) \int_a^b$$

$$\int_a^b w(x) Q(x) dx - \sum_{k=0}^m A_k Q(x_k) = \frac{1}{a_{m+1}} \int_a^b P_{m+1}(x) Q_{m-m-1}(x) w(x) dx = 0.$$

$m-m-1 \leq m$
 $m \leq 2m+1$

$$\Rightarrow R(Q) = 0, \forall Q \in \Pi_{2m+1}$$

Arătăm că polinomul ortogonal e unic:

$$P_p \neq P_{m+1}(x), U_{m+1}(x)$$

$$P_{m+1}(x) = a_{m+1} x^{m+1} + \dots$$

$$U_{m+1}(x) = b_{m+1} x^{m+1} + \dots$$

$$\Rightarrow H(x) = b_{m+1} P_{m+1}(x) - a_{m+1} U_{m+1}(x) \text{ este un polinom de grad cel mult } m$$

$$\Rightarrow H \in \Pi_m$$

$$\int_a^b H(x) x^k w(x) dx = b_{m+1} \underbrace{\int_a^b x^k P_{m+1}(x) w(x) dx}_{=0} - a_{m+1} \underbrace{\int_a^b x^k U_{m+1}(x) w(x) dx}_{=0}$$

$$\Rightarrow \int_a^b H(x) x^k w(x) dx = 0, k = \overline{0, m} \Rightarrow \text{polinomul are } m+1 \text{ răd } \in \mathbb{R}, \text{ distincte}$$

$$\Rightarrow H=0 \Rightarrow b_{m+1} P_{m+1}(x) = a_{m+1} U_{m+1}(x)$$

$\exists y_0, \dots, y_m, H(y_i) = 0$

$$\Leftrightarrow P_{m+1} = \lambda U_{m+1}, \lambda = \frac{a_{m+1}}{b_{m+1}}$$



Cazuri particulare.

Formula dreptunghiului. Formula dreptunghiurilor.

$$\int_a^b f(x) dx$$

$$P_1(x) = x - \frac{a+b}{2}$$

$$\int_a^b \left(x - \frac{a+b}{2}\right) dx = 0$$

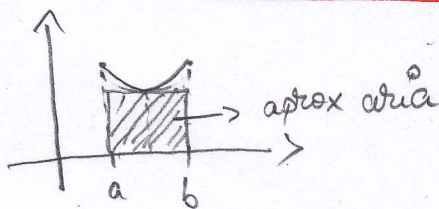
$$\left(\frac{x - \frac{a+b}{2}}{2}\right) \Big|_a^b = 0.$$

$$\int_a^b f(x) dx = A_0 f\left(\frac{a+b}{2}\right) + R(f)$$

$$f=1 \Rightarrow b-a = A_0$$

$$\int_a^b f(x) dx = (b-a) f\left(\frac{a+b}{2}\right) + R(f)$$

f. dreptunghiului



- Evaluăm restul $R(f)$

$$f(x) - f\left(\frac{a+b}{2}\right) \rightarrow \text{polim Lagrange de grad 1}$$

$$f(x) - f\left(\frac{a+b}{2}\right) = \left(x - \frac{a+b}{2}\right) [x, \frac{a+b}{2}; f] \Big|_{x = \frac{a+b}{2}} \Big|_a^b$$

$$\int_a^b \left(x - \frac{a+b}{2}\right) f(x) dx = \int_a^b \left(x - \frac{a+b}{2}\right)^2 [x, \frac{a+b}{2}; f] dx \equiv$$

aplicăm form. de medie pt fct pozitivă $(x - \frac{a+b}{2})^2$ și f . cont. $[x, \frac{a+b}{2}; f]$

$$\equiv \left[c, \frac{a+b}{2}; f\right] \int_a^b \left(x - \frac{a+b}{2}\right)^2 dx = \left[c, \frac{a+b}{2}; f\right] \frac{\left(x - \frac{a+b}{2}\right)^3}{3} \Big|_a^b$$

$$= \left[c, \frac{a+b}{2}; f\right] \left[\frac{(b-a)^3}{24} + \frac{(b-a)^3}{24}\right] = \frac{(b-a)^3}{12} \left[c, \frac{a+b}{2}; f\right]$$

$$\int_a^b \left(x - \frac{a+b}{2}\right) f(x) dx = \frac{(b-a)^3}{12} \left[c, \frac{a+b}{2}; f\right]$$

$$\begin{aligned} \int_a^b f(x) dx - (b-a) f\left(\frac{a+b}{2}\right) &= \int_a^b \left(x - \frac{a+b}{2}\right) [x, \frac{a+b}{2}; f] dx = \frac{(b-a)^3}{12} \left[c, \frac{a+b}{2}; [x, \frac{a+b}{2}; f]\right] \\ &= \frac{(b-a)^3}{12} [x, \frac{a+b}{2}; f]'(c_2) \end{aligned}$$

$$[x, \frac{a+b}{2}; f] = \frac{f(x) - f(\frac{a+b}{2})}{x - \frac{a+b}{2}} = \frac{\int_{\frac{a+b}{2}}^x f'(t) dt}{x - \frac{a+b}{2}} \quad \textcircled{=}$$

substituție

$$\frac{t - \frac{a+b}{2}}{x - \frac{a+b}{2}} = y \Rightarrow t = \frac{a+b}{2} + y(x - \frac{a+b}{2}) \Rightarrow dt = (x - \frac{a+b}{2}) dy.$$

$$\textcircled{=} \int_0^1 f'[\frac{a+b}{2} + y(x - \frac{a+b}{2})] dy.$$

$$[x, \frac{a+b}{2}; f]' = \int_0^1 y f''[\frac{a+b}{2} + y(x - \frac{a+b}{2})] dy \quad \text{form medie} \quad \frac{1}{2} f''[\frac{a+b}{2} + \theta(x - \frac{a+b}{2})]$$

$$\int_a^b f(x) dx - (b-a)f(\frac{a+b}{2}) = \frac{(b-a)^3}{24} \cdot f''(c_3) \quad c_3 = \frac{a+b}{2} + \theta(c_1 - \frac{a+b}{2})$$

Form. dreptunghiului cu rest

Pt f . dreptunghiurilor:

$$x_k = a + \frac{b-a}{n} k, \quad k=0, \overline{m} \quad (\text{am împărțit interval } [a,b] \text{ în } m \text{ intervale})$$

$$\int_a^b f(x) dx = \sum_{k=0}^{m-1} \int_{x_k}^{x_{k+1}} f(x) dx = \sum_{k=0}^{m-1} \left[(x_{k+1} - x_k) f\left(\frac{x_k + x_{k+1}}{2}\right) + \frac{(x_{k+1} - x_k)^3}{24} f''(\theta_k) \right] =$$

($f \in C''[a,b]$)

$$= \frac{b-a}{n} \sum_{k=0}^{m-1} f\left(\frac{x_k + x_{k+1}}{2}\right) + \frac{1}{24} \frac{(b-a)^3}{n^3} \cdot m \sum_{k=0}^{m-1} \frac{f''(\theta_k)}{n}$$

$$m_2 \leq \frac{\sum_{k=0}^{m-1} f''(\theta_k)}{n} \leq M_2 \quad (\text{prop. Darboux})$$

$$\Rightarrow \exists \theta \in [a,b] \text{ a.î. } f''(\theta) = \frac{\sum_{k=0}^{m-1} f''(\theta_k)}{n}$$

$$\int_a^b f(x) dx = \frac{b-a}{n} \sum_{k=0}^{m-1} f\left(\frac{x_k + x_{k+1}}{2}\right) + \frac{(b-a)^3}{24n^2} f''(\theta)$$

Form. dreptunghiurilor.

$$\text{ex 1: } \lim_{n \rightarrow \infty} \left[\int_a^b f(x) dx - \frac{b-a}{n} \sum_{k=0}^{m-1} f\left(\frac{x_k + x_{k+1}}{2}\right) \right] = 0.$$

$$\text{ex 2: } |f''| \leq M \Rightarrow \text{poate det cel mai mic } m \text{ pt } \varepsilon \text{ anum. aprox a int.}$$

$$\frac{(b-a)^3}{24n^2} M \leq \varepsilon \Rightarrow M \geq \frac{b-a \sqrt{b-a}}{\varepsilon \sqrt{24} \sqrt{M}}$$

- $w(x) = 1$ $[a, b]$ Polinoame Legendre.

$$P_m(x) = k_m \frac{d^m}{dx^m} [(x-a)^m (b-x)^m]$$

- $w(x) = \frac{1}{\sqrt{1-x^2}}$, $x \in [-1, 1]$ $T_m(x) = k_m \cos[m(\arccos x)]$

↓
pondera lui Chebâșev

↓
ne pot găsi toate răd polinomului
lui Chebâșev

$$m \arccos x = \frac{(2k+1)\pi}{2}$$

$$\arccos x = \frac{(2k+1)\pi}{2m} \quad x_k = \cos \frac{(2k+1)\pi}{2m}, \quad k = \overline{0, m-1}$$

! $A_k = \text{coef form de cuadratură}$ (pt examen)

[I] Coeficienții unei formule de cuadratură de tip Gauss sunt pozitivi:

$$\int_a^b f(x) w(x) dx = \sum_{k=0}^m A_k f(x_k) + R(f) \quad (\text{form. lui Gauss})$$

dem $A_k > 0$

$$A_k = \int_a^b l_k(x) w(x) dx$$

$$l_k^2(x) \quad R(l_k^2) = 0$$

$$\int_a^b \underbrace{l_k^2(x)}_{>0} \underbrace{w(x)}_{>0} dx = \sum A_k \delta_{k,k} = A_k. \Rightarrow \boxed{A_k > 0}, \forall k = \overline{0, m}$$

când $A_k > 0$

$$\sum_{k=0}^m |A_k| = \sum_{k=0}^m A_k = \int_a^b w(x) dx$$

Formula trapezului. Formula trapezelor

Fie $f: [a, b] \rightarrow \mathbb{R}$

Def. Sm. formula trapezului, formula de cuadratura

$\int_a^b f(x) dx = A_0 f(a) + A_1 f(b) + R(f)$ unde A_0, A_1 sunt a.i. formula de cuadratură să aibă grad de exactitate 1.

$$A_0 = \int_a^b \frac{x-b}{a-b} dx = \frac{(x-b)^2}{2(a-b)} \Big|_a^b = \frac{b-a}{2}$$

$$A_1 = \int_a^b \frac{x-a}{b-a} dx = \frac{b-a}{2}$$

$$\int_a^b f(x) dx = \frac{b-a}{2} [f(a) + f(b)] + R(f) \quad \text{Formula trapezului}$$

$$R_1 = R(x) = 0$$

$$R(x^3) = \frac{b^3-a^3}{3} - \frac{(b-a)(b^2+ab+a^2)}{2} = \frac{(b-a)(2b^2+2ab+2a^2-3b^2-3a^2)}{6} = \\ = \frac{(b-a)(-b^2-a^2+2ab)}{6} = \frac{(b-a)^3}{6}$$

$$f(x) = L_1(f; a, b)(x) = (x-a)(x-b)[x, a, b; f]$$

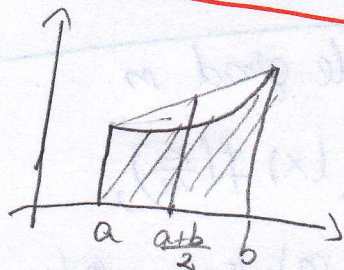
$$\int_a^b f(x) dx - \frac{b-a}{2} [f(a) + f(b)] = \int_a^b \underbrace{(x-a)(b-x)}_{f. \text{ medie}} [x, a, b; f] dx = \\ = -[c, a, b; f] \int_a^b (x-a)(b-x) dx$$

$$\int_a^b (x-a)(b-x) dx = (b-a) \frac{(b-a)^2}{2} - \frac{(b-a)^3}{3} = \frac{(b-a)^3}{6}$$

$$\int_a^b f(x) dx = \frac{f(a)+f(b)}{2} (b-a) - \frac{(b-a)^3}{6} f''(\theta)$$

$$\int_a^b f(x) dx = \frac{f(a)+f(b)}{2} (b-a) - \frac{(b-a)^3}{12} f''(\theta)$$

Form trapezului
cu rest.



Obs Dacă f este o fct convexă atunci are loc dubla inegalitate

$$(b-a)f\left(\frac{a+b}{2}\right) \leq \int_a^b f(x) dx \leq \frac{f(a)+f(b)}{2} (b-a) \quad \text{Hadamard}$$

$$f: [a, b] \rightarrow \mathbb{R} \quad x_k = a + \frac{b-a}{n} k, \quad k=0, \dots, m$$

$$\int_a^b f(x) dx = \sum_{k=0}^{m-1} \int_{x_k}^{x_{k+1}} f(x) dx \quad \text{---}$$

pe fiecare int $[x_k, x_{k+1}]$ aplic. f. trapezului

$$\ominus \sum_{k=0}^{m-1} \frac{(x_{k+1} - x_k)}{2} [f(x_k) + f(x_{k+1})] - \frac{(x_{k+1} - x_k)^3}{12} f''(\theta_k) =$$

$$= \frac{b-a}{2m} \sum_{k=0}^{m-1} (f(x_k) + f(x_{k+1})) - \frac{(b-a)^3}{12m^2} \sum_{k=0}^{m-1} f''(\theta_k) =$$

$$= \frac{b-a}{2m} \left[f(a) + 2 \sum_{k=1}^{m-1} f(x_k) + f(b) \right] - \frac{(b-a)^3}{12m^2} f''(\theta)$$

$$\int_a^b f(x) dx = \frac{b-a}{2m} \left[f(a) + 2 \sum_{k=1}^{m-1} f(x_k) + f(b) \right] - \frac{(b-a)^3}{12m^2} f''(\theta)$$

Formula trapezelor.

Aproximarea uniformă a funcțiilor

T. Weierstrass

Dacă $f: [a, b] \rightarrow \mathbb{R}$ continuă, atunci $\exists P_n$ astfel încât $\|f - P_n\| \xrightarrow{n \rightarrow \infty} 0$

S.N. Bernstein $\rightarrow$ dem. constructivă

Def. Fie $f: [0, 1] \rightarrow \mathbb{R}$. S.m. polinom Bernstein de grad m

$$B_m(f)(x) = \sum_{k=0}^m \binom{m}{k} x^k (1-x)^{m-k} f\left(\frac{k}{m}\right) = \sum_{k=0}^m p_{m,k}(x) f\left(\frac{k}{m}\right),$$

$$p_{m,k}(x) = \binom{m}{k} x^k (1-x)^{m-k}$$

$\square$ $\lim_{m \rightarrow \infty} \|f - B_m(f)\| = 0, \forall f \text{ et continua } f: [0, 1]$

Def. Fie $f: [0,1] \rightarrow \mathbb{R}$. S.m modul de continuitate al funcției f .
Fct $\omega: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ definită prin $\omega(f, \delta) = \sup_{|x-y| \leq \delta} |f(x) - f(y)|$

$$\omega(f, 1) = M - m$$

$$\delta \geq 1 \Rightarrow \omega(f, \delta) = M - m$$

Obs $\omega(f, \delta)$ funcție crescătoare în raport cu δ

$$A \subset B \Rightarrow \sup A \leq \sup B$$

$$\delta_1 < \delta_2 \Rightarrow \omega(f, \delta_1) \leq \omega(f, \delta_2)$$

$$\Rightarrow \omega(f, \delta_1) \leq \omega(f, \delta_2)$$

□. Fct f este continuă dacă și numai dacă $\delta = 0$ și $\omega(f, \delta) = 0$

$\forall \varepsilon > 0, \exists \delta > 0$. a.î. $\forall$ de câte ori $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

$$\Rightarrow \sup_{|x-y| \leq \delta} |f(x) - f(y)| \leq \varepsilon$$

$$\delta_m \rightarrow 0$$

$$\sup_{|x-y| \leq \delta_m} |f(x) - f(y)| \leq \varepsilon$$

$$\varepsilon \rightarrow 0$$

$$0 \leq \lim_{n \rightarrow \infty} \omega(f, \delta_n) \leq 0$$